

Academic Stress and Mental Health Among School Students: The Mediating Role of Depression

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Abstract

This study examines the relationship between academic stress and mental health among school students, with a particular focus on the mediating role of depression. In today's competitive educational environment, students are increasingly exposed to academic pressures that may negatively affect their psychological well-being.

The study is based on primary data collected through a structured questionnaire using a 5-point Likert scale. A quantitative research approach was adopted, and statistical techniques including correlation analysis, regression analysis, and mediation analysis were used to examine the relationships among variables.

The findings indicate that academic stress significantly affects both depression and mental health. Depression also has a strong influence on mental health and partially mediates the relationship between academic stress and mental health. The results suggest that higher academic stress leads to increased depressive symptoms, which in turn contribute to poorer mental health outcomes.

Overall, the study highlights the importance of managing academic stress and addressing depression to improve students' mental well-being. It also provides practical insights for educators and policymakers to design interventions aimed at reducing psychological distress among students.

Keywords: Academic Stress, Depression, Mental Health, Mediation, Students

Introduction

Adolescence is a critical developmental stage characterized by significant psychological, emotional, and social changes. During this period, school environments play a central role in shaping students' well-being. However, alongside developmental opportunities, students are increasingly exposed to academic demands that contribute to heightened stress levels. Research indicates that students frequently experience stress related to examinations, performance expectations, and fear of failure, making academic environments a major source of psychological pressure (Pascoe et al., 2020).

In contemporary education systems, particularly those driven by competition and performance-based evaluation, academic stress has become a dominant concern. Students are required not only to meet institutional expectations but also to satisfy parental and societal demands. Evidence suggests that pressures arising from workload, examinations, and expectations significantly contribute to students' psychological distress (Deb et al., 2015). Over time, persistent exposure to such stressors may impair coping abilities and increase vulnerability to mental health problems.

One of the most significant consequences of academic stress is its association with depression. Depression is characterized by persistent sadness, loss of interest, reduced energy, and impaired concentration. Studies have demonstrated that academic stress is positively associated with depressive symptoms, particularly when students perceive a gap between expectations and achievement (Liu & Lu, 2011). This suggests that depression may function as an important psychological mechanism through which academic stress influences broader mental health outcomes.

Mental health among students is commonly assessed through indicators such as anxiety and stress, which reflect emotional and psychological well-being. Research has shown that increased levels of stress are strongly associated with a higher likelihood of mental disorders,

including anxiety and mood-related conditions (Karyotaki et al., 2020). Additionally, prevalence studies reveal that a substantial proportion of students experience moderate to severe levels of depression, anxiety, and stress, highlighting the seriousness of this issue (Ramón-Arbués et al., 2020).

To better understand these relationships, theoretical frameworks such as the demands–resources model provide valuable insights. This model suggests that excessive demands, such as academic pressure, lead to negative psychological outcomes, including burnout and depressive symptoms, particularly when adequate resources are lacking (Salmela-Aro & Upadaya, 2014). Furthermore, research on coping and emotion regulation indicates that individuals' ability to manage stress plays a crucial role in determining whether stress leads to adverse psychological outcomes (Compas et al., 2017).

Based on this perspective, the present study focuses on three key variables: academic stress (independent variable), depression (mediating variable), and mental health—measured through anxiety and stress (dependent variable). By examining the relationships among these variables, the study aims to provide a clearer understanding of how academic stress influences students' mental health through the mediating role of depression.

Literature Review

Overview of Existing Literature

Mental health issues among students have gained increasing attention in recent years, particularly during adolescence when individuals are more susceptible to psychological distress. Existing literature highlights that students experience multiple stressors, including academic, social, and personal pressures, which contribute to outcomes such as anxiety, depression, and stress. Although a substantial body of research has focused on higher education populations, fewer studies have explored these relationships among school-level students, indicating a gap that warrants further investigation.

Academic Stress

Academic stress is widely recognized as a major factor influencing students' psychological well-being. It arises from various academic demands, including examinations, workload,

competition, and expectations from parents and teachers. Studies have shown that a large proportion of students report stress related to academic performance, with many experiencing anxiety about tests and fear of poor grades (Pascoe et al., 2020).

Parental expectations are also a significant contributor to academic stress. Research indicates that many students perceive strong pressure from parents to achieve high academic performance, which increases their stress levels (Deb et al., 2015). Furthermore, the Academic Expectations Stress Inventory identifies expectations from parents, teachers, and self as key dimensions of academic stress, emphasizing that stress is influenced not only by workload but also by perceived expectations (Ang & Huan, 2006). These findings suggest that academic stress is a multidimensional construct that significantly impacts students' mental health.

Depression

Depression is one of the most common psychological outcomes associated with prolonged exposure to stress. Among students, depressive symptoms include persistent sadness, lack of interest in activities, low energy, and difficulty concentrating. Empirical studies have demonstrated a strong positive relationship between academic stress and depression, indicating that students experiencing higher levels of academic stress are more likely to develop depressive symptoms (Liu & Lu, 2011).

In addition, depression plays a crucial mediating role in the relationship between stress and mental health. Research suggests that stress-related experiences can lead to emotional disturbances, which, if not effectively managed, result in depressive symptoms that further impact overall psychological well-being (Ramón-Arbués et al., 2020). Therefore, depression is not only an outcome of academic stress but also an important mechanism through which stress affects mental health.

Mental Health (Anxiety and Stress)

Mental health in students is often conceptualized through indicators such as anxiety and stress. Anxiety refers to persistent feelings of worry and nervousness, while stress reflects the body's response to challenging situations. Studies have shown that a significant proportion of students

experience symptoms of anxiety and stress, often at moderate to severe levels (Bayram & Bilgel, 2008).

Furthermore, research indicates a strong association between perceived stress and mental disorders. Higher levels of stress are linked to an increased likelihood of developing anxiety and depressive disorders, demonstrating the critical role of stress in shaping mental health outcomes (Karyotaki et al., 2020). These findings highlight the importance of understanding the factors that contribute to students' mental health.

Theoretical Framework

The relationships among academic stress, depression, and mental health can be explained through the demands–resources model. This model proposes that high demands, such as academic workload and expectations, lead to negative psychological outcomes when individuals lack sufficient resources to cope effectively (Salmela-Aro & Upadyaya, 2014). Specifically, excessive academic demands can result in emotional exhaustion and depressive symptoms, which subsequently affect overall well-being.

Additionally, coping and emotion regulation theories suggest that individuals' responses to stress play a crucial role in determining psychological outcomes. Effective coping strategies and adaptive emotion regulation are associated with lower levels of psychological distress, whereas maladaptive coping strategies increase the risk of anxiety and depression (Compas et al., 2017). These theoretical perspectives support the idea that academic stress influences mental health both directly and indirectly through depression.

Conceptual Framework of the Study

Based on the existing literature, the present study proposes a conceptual framework in which academic stress serves as the independent variable, depression acts as a mediating variable, and mental health (anxiety and stress) functions as the dependent variable. Previous research supports the direct relationship between academic stress and depression, as well as the association between depression and mental health outcomes. Furthermore, theoretical models suggest that depression mediates the relationship between stress and psychological well-being.

This framework provides a structured understanding of how academic stress contributes to mental health outcomes and highlights the importance of addressing depression as a key mechanism in this relationship.

Research Gap

Despite the growing body of research, several gaps remain. First, many studies focus on university students or general stress factors, with limited emphasis on school-level populations. Second, while academic stress and mental health have been widely studied, fewer studies have examined the mediating role of depression in this relationship among school students. Third, there is a lack of context-specific research in regions such as Varanasi, where socio-cultural and educational pressures may differ significantly. Therefore, the present study aims to address these gaps by examining the influence of academic stress on the mental health of school students, with a specific focus on the mediating role of depression.

Conceptual Framework

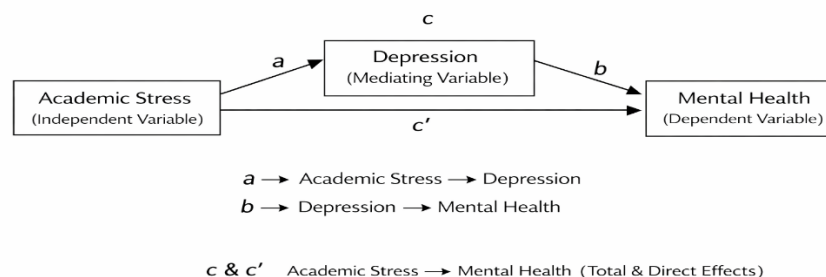


Figure 1

Conceptual Framework of Academic Stress, Depression, and Mental Health

Research Objectives

1. To assess the level of academic stress among school students in Varanasi.
2. To evaluate the mental health status of students in terms of:

- Anxiety
 - Depression
 - Stress
3. To analyze the relationship between academic stress and mental health outcomes among students.
 4. To identify key factors contributing to academic stress (e.g., exams, parental pressure, workload).
 5. To examine whether academic stress significantly predicts poor mental health among students.
 6. To suggest practical interventions for schools to reduce academic stress and improve student mental well-being.

Hypotheses

H1: Academic stress has a significant effect on mental health.

H2: Academic stress has a significant effect on depression.

H3: Depression has a significant effect on mental health.

H4: Depression mediates the relationship between academic stress and mental health.

Research Methodology

Research Design

This study adopts a quantitative research design to examine the relationship between academic stress, depression, and mental health among school students. A causal research approach is used to identify the effect of academic stress on mental health, as well as the mediating role of depression.

Data Collection Method

The study is based on primary data collected through a structured questionnaire distributed using Google Forms. The questionnaire included items measuring academic stress, depression, and mental health using a 5-point Likert scale ranging from strongly disagree to strongly agree.

Sampling Technique and Sample Size

A non-probability convenience sampling method was used due to accessibility constraints. The final sample consists of **89 respondents**, representing school students.

Measurement of Variables

- **Academic Stress (IV):** Measured using items related to exams, workload, parental pressure, and competition
- **Depression (MV):** Measured using emotional and psychological indicators
- **Mental Health (DV):** Measured through anxiety and stress-related indicators

Data Analysis Technique

The data were analyzed using Python-based statistical tools. The analysis includes:

- Reliability analysis (Cronbach's Alpha)
- Correlation analysis
- Regression analysis
- Mediation analysis
- Multiple regression

This approach ensures accuracy, reproducibility, and robust statistical interpretation.

Analysis and Discussion

Demographic Profile of Respondents

Descriptive Statistics

Variable	N	Mean	SD	Min	Max
Academic Stress	89	3.83	0.77	1.00	5.00
Depression	89	3.55	0.90	1.60	5.00
Mental Health	89	3.52	0.82	1.57	5.00

Table 1: Descriptive Statistics

The results indicate that students experience moderate to high levels of academic stress. Similarly, depression and mental health scores suggest a noticeable presence of psychological distress among students. This reflects the growing concern regarding student mental well-being in academic environments.

Reliability Analysis

Variable	Cronbach’s Alpha
Academic Stress	0.75
Depression	0.81
Mental Health	0.86

Table 2

Reliability Analysis (Cronbach’s Alpha)

All constructs show acceptable to high internal consistency ($\alpha > 0.70$), confirming that the measurement scales are reliable and suitable for further analysis.

Correlation Analysis

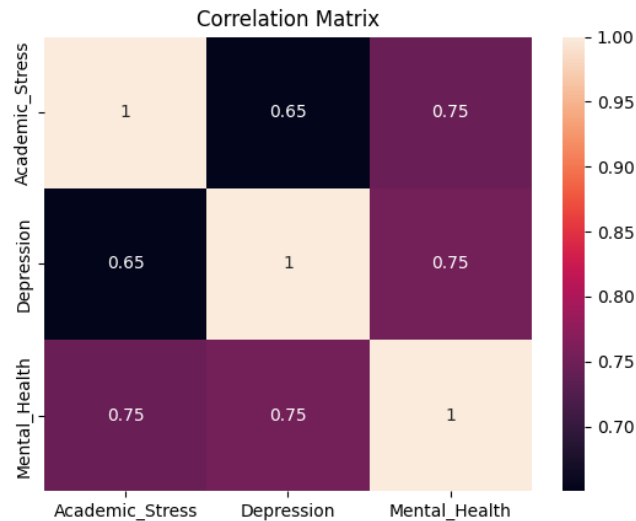


Figure 2
Correlation Matrix Heatmap

The correlation results indicate strong positive relationships among all variables. Academic stress is strongly associated with depression and mental health issues, suggesting that increased stress contributes significantly to psychological distress. The strong relationship between depression and mental health further supports its mediating role.

Regression Analysis

H1: Academic Stress → Mental Health

Dependent Variable: Mental Health

Predictor	B	p-value	R ²
Academic Stress	0.79	< .001	0.556

Academic stress has a significant positive effect on mental health ($\beta = 0.79$, $p < .001$). Higher academic stress leads to poorer mental health.

H2: Academic Stress → Depression

Dependent Variable: Depression

Predictor	B	p-value	R ²
Academic Stress	0.76	< .001	0.423

Academic stress significantly predicts depression ($\beta = 0.76$, $p < .001$).
Increased stress results in higher depressive symptoms.

H3: Depression → Mental Health

Dependent Variable: Mental Health

Predictor	B	p-value	R ²
Depression	0.68	< .001	0.568

Depression significantly affects mental health ($\beta = 0.68$, $p < .001$).
Higher depression worsens mental health outcomes.

Mediation Analysis (H4)

Effect Type	Coefficient	p-value	Result
Total Effect	0.79	< .001	Significant
Direct Effect	0.47	< .001	Significant
Indirect Effect	0.32	< .001	Significant

The mediation analysis shows that:

- Direct effect = Significant
- Indirect effect = Significant
- Type = **Partial Mediation**

Academic stress affects mental health both directly and indirectly through depression.

Multiple Regression Analysis

Predictor	B	p-value
Academic Stress	0.47	< .001

Depression	0.42	< .001
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Model Fit: $R^2 = 0.681$

Both academic stress and depression significantly predict mental health. The model explains **68.1% variance ($R^2 = 0.681$)**, indicating strong explanatory power.

Scatter Plot Analysis

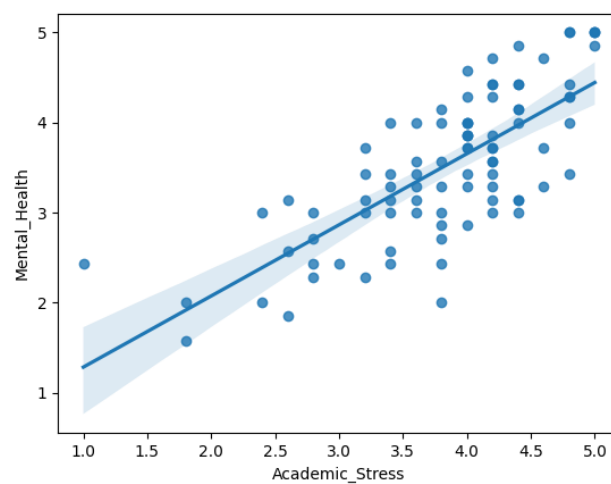


Figure 3
Academic Stress → Mental Health

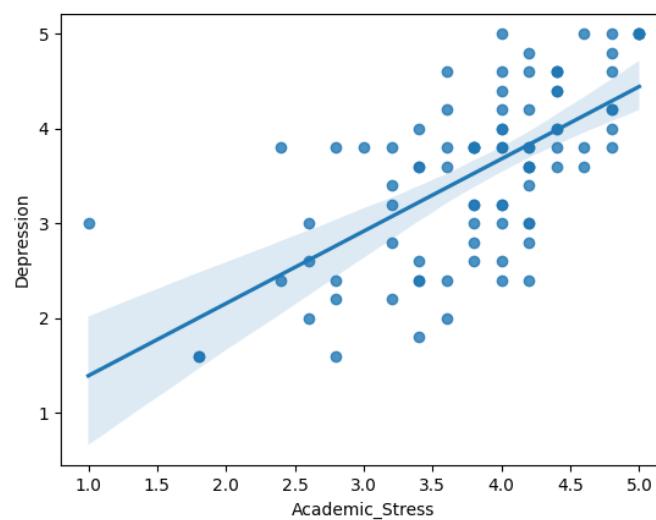


Figure 4
Academic Stress → Depression

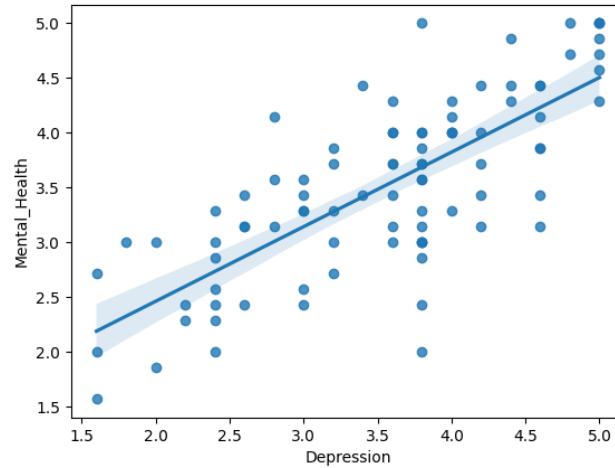


Figure 5
Depression → Mental Health

All scatter plots show clear positive linear relationships, confirming that increases in academic stress and depression are associated with worsening mental health outcomes.

Findings

- Academic stress significantly impacts mental health
- Academic stress increases depression
- Depression strongly affects mental health
- Depression partially mediates the relationship
- Students experience moderate to high psychological distress

Implications

From a theoretical perspective, this study supports the demands–resources model by demonstrating how excessive academic demands lead to negative psychological outcomes.

From a practical perspective:

- Schools should reduce academic pressure
- Mental health awareness programs are necessary
- Counseling and stress management strategies should be implemented

Conclusion

The study concludes that academic stress plays a crucial role in influencing students' mental health. Depression acts as a key mechanism through which stress affects psychological well-being. While academic pressure is unavoidable, its negative effects can be minimized through proper interventions and support systems.

Limitations

- Limited sample size (89 respondents)
- Based on self-reported data
- Restricted to a specific region

Future research can include larger and more diverse samples.

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